



The Tale of COVID-19's Herd Immunity

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Introduction

The COVID-19 pandemic, sparked by the novel coronavirus SARS-CoV-2, has left an indelible mark on the world, including the United States of America. Beginning in December 2019 in Wuhan, China, the virus swiftly evolved into a global pandemic.

With a population of approximately 335 million, the United States experienced significant COVID-19 impacts. A staggering 26.19% of the population fell victim to the virus. However, the nation's efforts in vaccination proved noteworthy, with 80.71% of the population receiving vaccines. Despite these efforts, there was tragic loss of life, accounting for 0.25% of the population.

This report delves deeper into the data surrounding COVID-19's impact on the United States, aiming to provide insights into the virus's spread, vaccination efforts, and the toll it took on the nation. By analyzing these statistics, the challenges faced and the progress made in combating this unprecedented global health crisis is better comprehensible.

Virus-Related Deaths Analysis (2020-2023)

This dashboard presents a comprehensive analysis of virus-related deaths from 2020 to 2023, focusing on COVID-19, Pneumonia, and Influenza. The following key findings are derived from the provided data:



Figure 1: Virus Death Rate in Relation to Pneumonia & Influenza

The data reveals a dynamic pattern of virus-related deaths over these years. Notably, COVID-19 displayed fluctuations in its impact, with the highest toll seen in 2021. Pneumonia also exhibited significant mortality, while Influenza-related deaths remained minimal throughout the analyzed period.

It's essential to continue monitoring and researching these trends to inform public health strategies and interventions effectively. Further investigation into the causes of variations in mortality rates among these viruses is warranted to ensure optimal healthcare preparedness and response.

2021

0.91 million **COVID-19** deaths
 0.81 million **Pneumonia** deaths
 0 **Influenza** deaths

2022

0.49 million **COVID-19** deaths
 0.53 million **Pneumonia** deaths
 0.02 million **Influenza** deaths

2023

0.07 million **COVID-19** deaths
 0.16 million **Pneumonia** deaths
 0.01 million **Influenza** deaths

Vaccine Distribution & Administered (2021 - 2023)

This visualization provides a concise overview of the distribution and administered of COVID-19 vaccines across the years 2021, 2022, and up until May 2023. The analysis focuses on key vaccine manufacturers, namely Pfizer, Moderna, Janssen, and Novovax, highlighting the distribution and vaccine administered figures for each year.

Vaccine Distribution and Administered Overview

In 2021, approximately 579 million COVID-19 vaccines were distributed, with Pfizer leading at 352.27 million, followed by Moderna at 221.14 million, and Janssen at 27.38 million. However, of these distributed vaccines, only around 483 million were administered. Pfizer accounted for 280.84 million doses administered, Moderna for 184.64 million, and Janssen for 16.98 million.



Figure 2: Vaccine Distribution & Administered

In 2022, vaccine distribution decreased to approximately 291 million doses. Pfizer distributed 175.62 million, Moderna 112 million, Janssen 2.88 million, and Novovax 0.61 million.

Vaccine administration in 2022 totaled around 137 million doses. Pfizer accounted for 87.131 million doses administered, Moderna for 48.19 million, Janssen for 1.51 million, and Novovax for 0.05 million.

In 2023, up to May, there were approximately 33 million vaccine doses distributed. Pfizer led with 95.13 million, followed by Moderna with 44.78 million, Janssen with 0.11 million, and Novovax with 0.29 million. Vaccine administration for the same period was approximately 10 million doses. Pfizer accounted for 38.86 million doses administered, Moderna for 21.82 million, Janssen for 0.05 million, and Novovax for 0.01 million.

[Vaccine Distribution and Administered Overview](#)

The year 2021 saw the highest distribution and administration of COVID-19 vaccines, with Pfizer being the most widely administered vaccine. In 2022, both distribution and administration declined, likely reflecting improved vaccination coverage. 2023 (up to May) showed a decrease in distribution but an increase in the number of vaccines administered compared to 2022, suggesting ongoing vaccination efforts.

[Difference in Distributed and Administered Vaccines](#)

In 2021, the difference between distributed and administered vaccines was approximately 96 million doses. In 2022, this difference decreased to around 154 million doses. In 2023 (up to May), the difference between distribution and administration was roughly 23 million doses.

The data presented in this report highlights the trends in COVID-19 vaccine distribution and administration over the years 2021 to 2023. While the highest vaccine distribution and administration occurred in 2021, efforts continue into 2023 to ensure vaccines reach those in need. Understanding these trends is crucial for assessing the progress of vaccination campaigns and optimizing future strategies.

Infection, Death and Vaccination

This chart examines the trends in COVID-19 vaccinations, deaths, and infection cases from January 2021 to May 2023. The aim is to identify correlations between the number of deaths and vaccinations, as well as between the number of deaths and COVID-19 infection cases. A noticeable pattern is the seasonal rise in death rates during winter months.

1. Correlation between Number of Deaths and Vaccinations

To assess the relationship between the number of deaths and COVID-19 vaccinations, the data across the provided timeline has been analyzed.

In 2021, the death count started relatively low in January but increased through the year, peaking in December at 46,610 deaths. Vaccinations began in January with 25,266,079 doses and consistently increased each month, peaking in December with 46,906,515 doses administered.

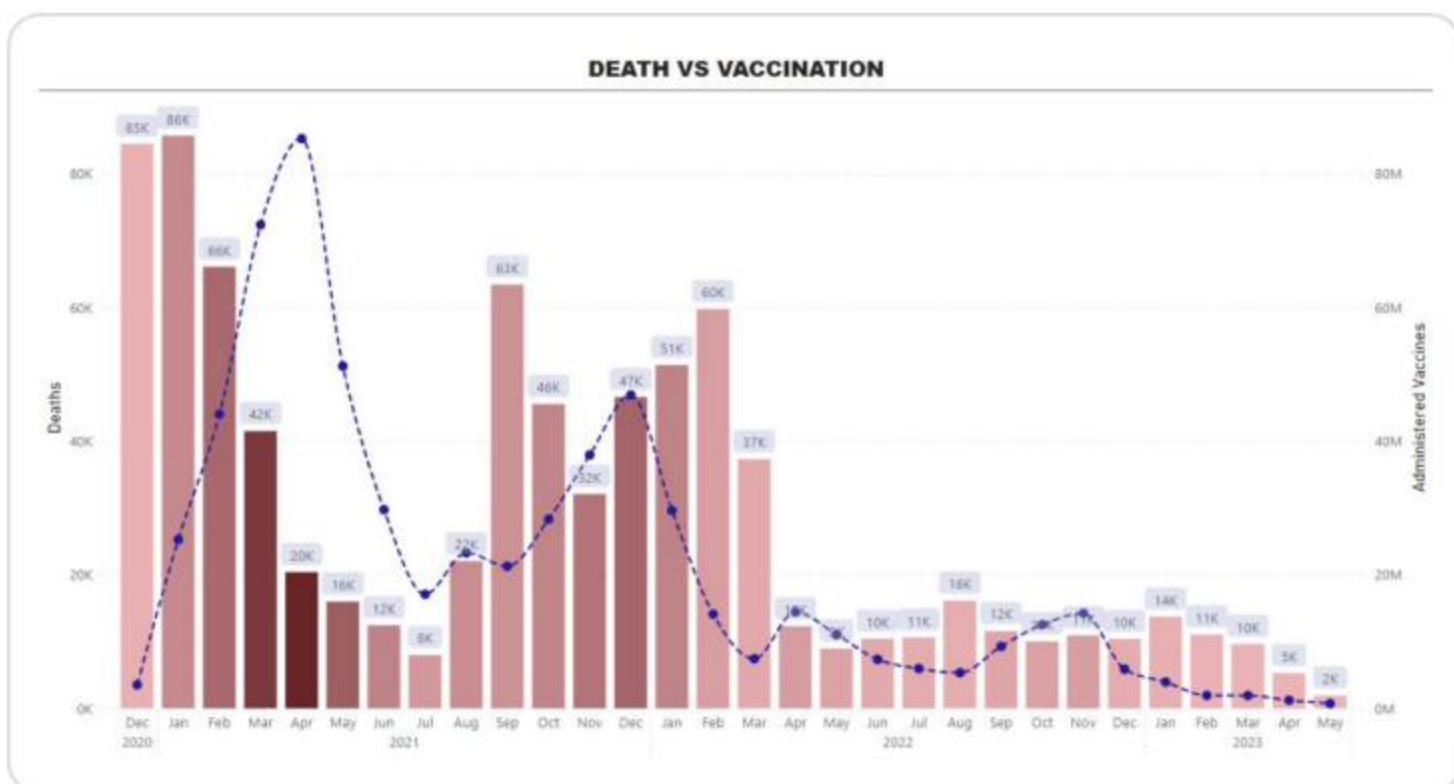


Figure 3: Correlation between Deaths and Vaccinations

In 2022, deaths and vaccinations both exhibited a decrease in the first half of the year. Deaths reached a low point in April with 12,308, while vaccinations decreased to 5,865,362 in December.

In 2023, both deaths and vaccinations continued their downward trend. Deaths reached 1,920 in May, while vaccinations were at 3,920,734 in January.

Correlation Analysis: There appears to be an inverse correlation between the number of deaths and the number of vaccinations administered. As vaccination rates increased, the number of deaths decreased, and vice versa.

2. Correlation between Number of Deaths and Infection

Now, the correlation between the number of deaths and COVID-19 infection cases can be examined.

In 2021, deaths reached their peak of 85,754 in January, coinciding with a high number of infection cases (5,632,865). Deaths gradually decreased throughout the year, while infection cases fluctuated but remained relatively high.

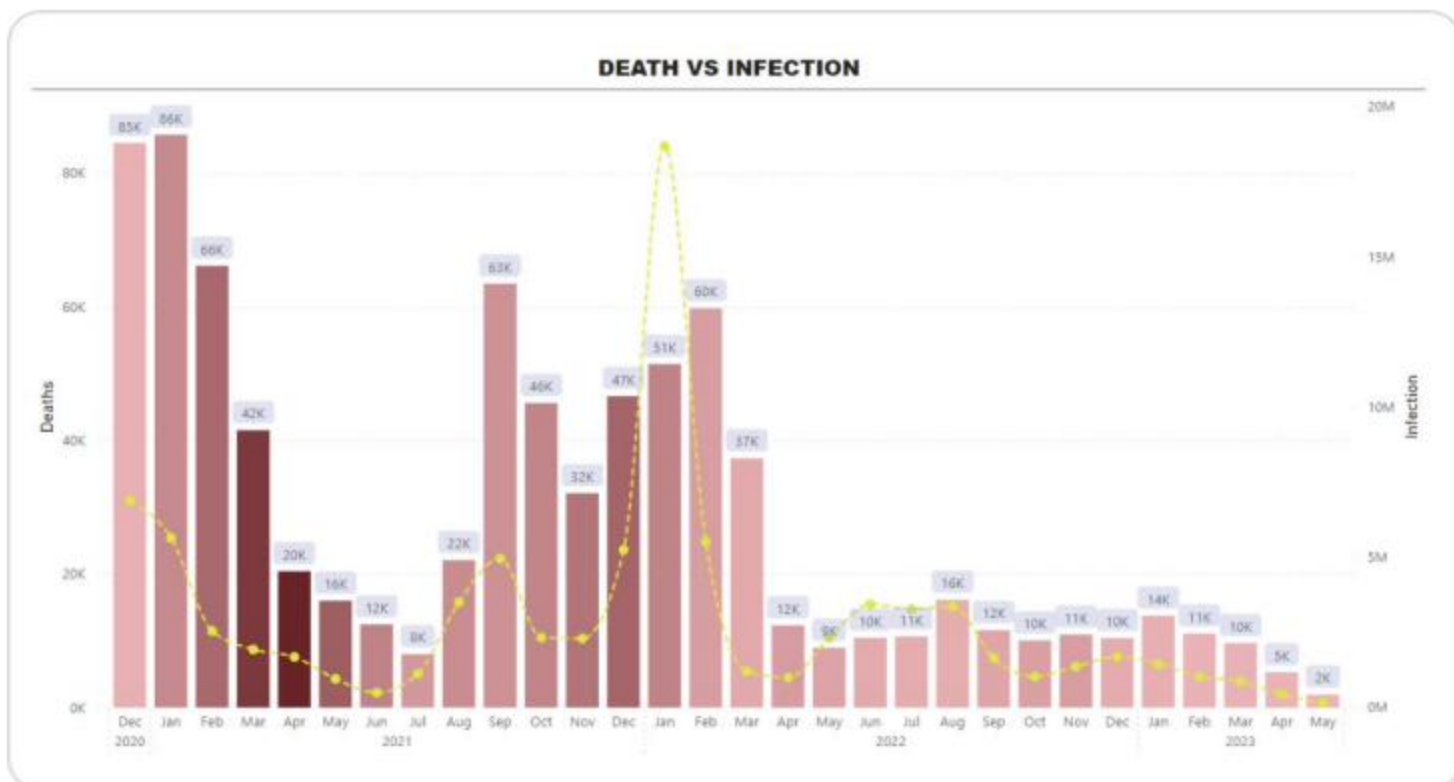


Figure 4: Correlation between Deaths and Infections

In 2022, both deaths and infection cases decreased significantly. Deaths reached their lowest point in April (12,308) while infection cases reached a low point in March (1,194,540).

In 2023, deaths and infection cases continued to decrease. Deaths reached 1,920 in May, while infection cases stood at 150,623.

Correlation Analysis: There is a plain correlation between the number of deaths and the number of COVID-19 infection cases. As infection cases increased, the number of deaths also tended to rise, and as infection cases decreased, so did the number of deaths.

This visualization highlights two key correlations related to the COVID-19 pandemic. First, there is an inverse correlation between the number of deaths and the number of vaccinations, indicating that increased vaccination rates are associated with decreased mortality. Second, there is a positive correlation between the number of deaths and the number of infection cases, indicating that higher infection rates are associated with higher mortality.

It is essential to consider these correlations in public health strategies, emphasizing the importance of vaccination to reduce mortality and continuing efforts to control and prevent the spread of the virus to minimize both cases and deaths. Additionally, the observed seasonal pattern of higher death rates during the winter months underscores the need for vigilance and preparedness during these periods.

Infection, Death, and Vaccination by State

This section consists of four map charts focusing on Population, Vaccination, Infection, and Death statistics. These charts offer valuable insights into the COVID-19 situation across different U.S. states.

Population

The population map chart clearly displays the population density of various states. California leads with a staggering population of 39 million, followed by approximately 30 million in Texas, 22 million in Florida, 19.5 million in New York, 13 million Pennsylvania, and 12.5 million in Illinois. These figures help provide context for understanding the impact of COVID-19 across the states.

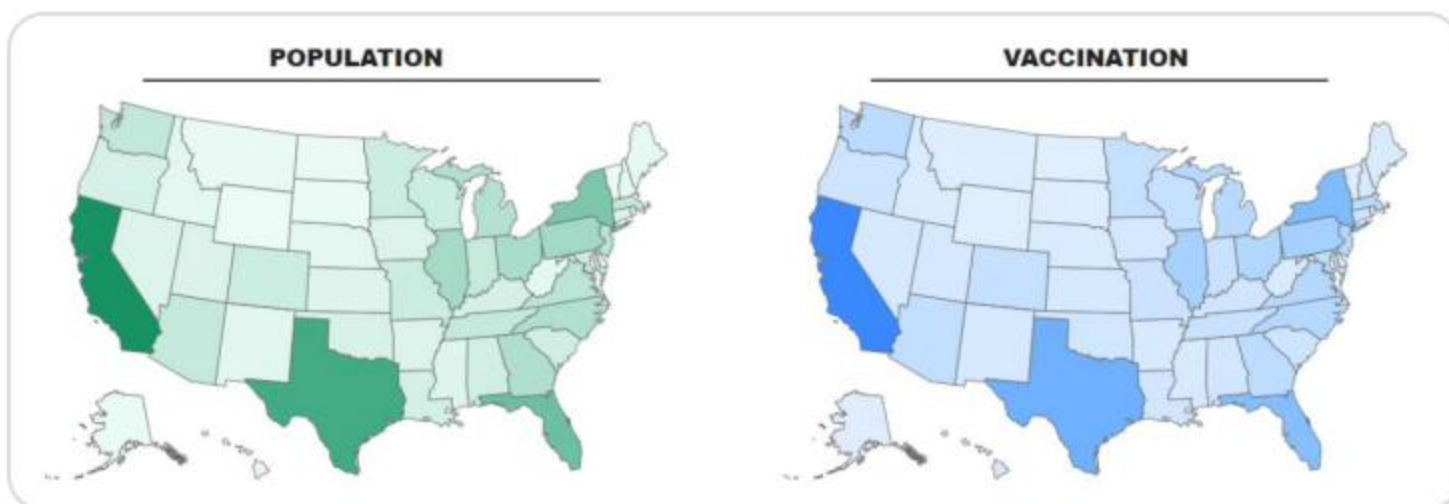


Figure 5: USA Population and Vaccination Density Map

Vaccination

In the vaccination map chart, the number of vaccinated individuals in each state's population is highlighted. California leads in vaccination efforts, with approximately 17.5 million doses administered. Texas follows closely with 15.5 million doses, then 14 million in New York, and 13.5 million in Florida. These statistics are vital for evaluating progress towards achieving herd immunity and understanding the correlation with population and infection data.

Infection

The infection map chart presents the total reported COVID-19 cases in each state. California reports the highest number of infections, with 11 million cases, followed by 7.3 million in Texas, 6.5 million in Florida, 3.4 million in Illinois, and 3.2 million in New York. The data enables an assessment of the pandemic's spread magnitude and its relationship with population and vaccination statistics.

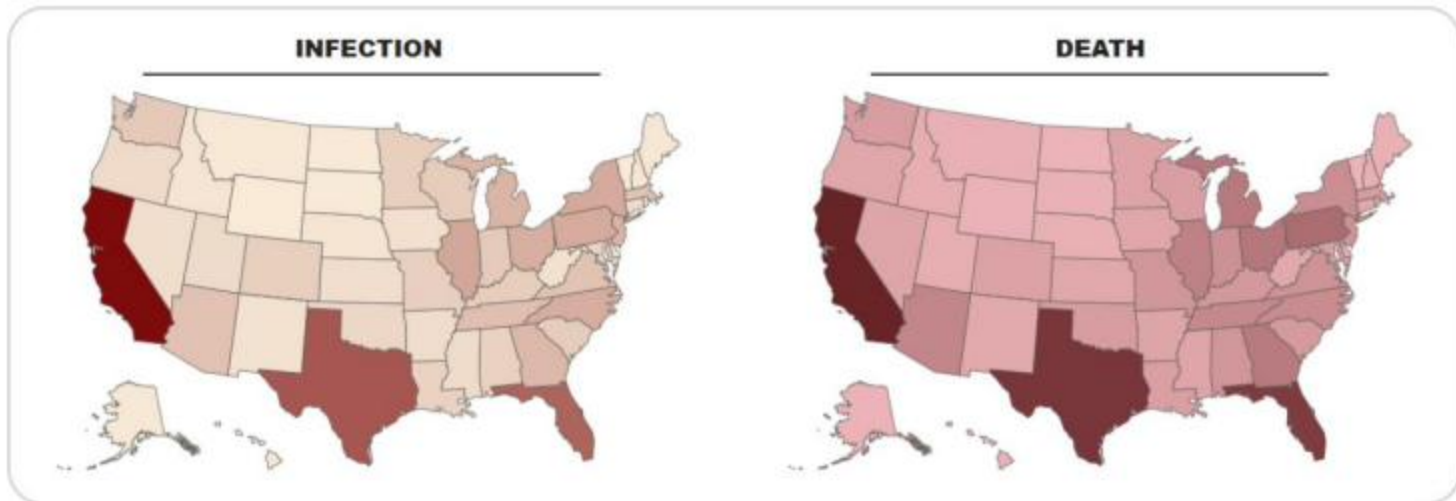


Figure 6: USA COVID-19 Infection and Death Density Map

Death

The death map chart visualizes the total COVID-19-related deaths in each state, from highest to lowest. California reports the highest number of deaths, with approximately 83 thousand, followed by 72 thousand in Texas and 68.9 thousand in Florida. This data provides insights into the severity of the pandemic's impact and its correlation with population, infection, and vaccination data.

Variants and Seasonal Trends

Introduction

The COVID-19 pandemic has been marked by the emergence and spread of various SARS-CoV-2 variants, each with its unique characteristics. This report aims to provide a data-driven analysis of the key observations regarding the evolution of COVID-19 variants, with a particular focus on the Delta, Omicron (BA.1), Omicron (BA.5), and a note of caution on Omicron (XBB.1.5) as an emerging variant. Additionally, we will explore any correlations between these variants and seasonal fluctuations.

Alpha Variant

The Alpha variant made its debut in December 2020, initially infecting a small number of individuals. However, by March 2021, it had become the dominant strain, infecting tens of thousands. Surprisingly, it dwindled rapidly after April 2021, with a peak of 89,647 cases in April, and was no longer detected by early 2022, indicating its relatively short-lived prevalence.

Beta Variant

The Beta variant emerged in January 2021, with a peak of 1,182 cases in April of that year. Subsequently, it exhibited a rapid decline, and no cases have been reported since September 2021.

Delta Variant

The Delta variant appeared in early 2021 and, unlike the Alpha and Beta variants, exhibited a sustained presence, reaching its high point in August 2021 with a staggering 393,387 cases. However, it showed a notable decline by early 2022, with only one case reported in February 2023, indicating a significant decrease in its prevalence.

Gamma Variant

The gamma variant's elevated transmissibility and substantial virulence led to higher mortality rates in most age groups. This variant had a fluctuating presence in 2021, peaking in November with 230,262 cases. Since then, no new cases of the Gamma variant have been reported.

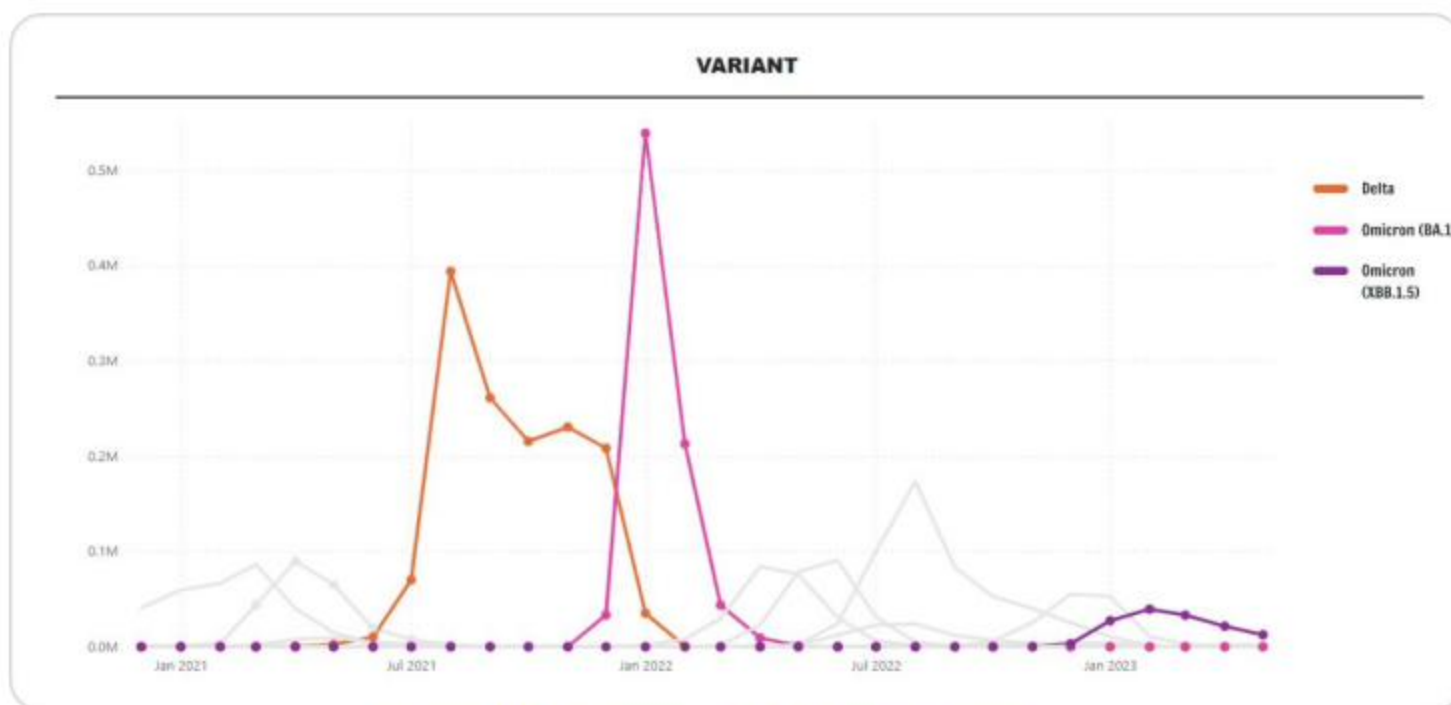


Figure 7: Infections caused by different Variants

Omicron Variants

Omicron is a highly concerning and prolific variant that has undergone several sub-lineages. Two prominent sub-lineages are Omicron (BA.1) and Omicron (BA.5), which demand special attention.

[Omicron \(BA.1\)](#) was first identified in October 2021 and rapidly became widespread. In early 2022, cases surged, peaking in January with over 538,301 cases reported. However, by May 2023, it has nearly disappeared, with only 32 cases reported.

[Omicron \(BA.5\)](#) emerged around the same time as Omicron (BA.1) and exhibited a similar pattern, albeit with a slightly delayed peak. It reached its highest number of cases in August 2022, with 173,245 cases reported, followed by a gradual decline.

Emerging Variant: Omicron (XBB.1.5)

Omicron (XBB.1.5) is a relatively new variant, first detected in November 2022. Alarming, it rapidly gained momentum and became a significant concern by December 2022, with 31,102 cases reported. In early 2023, it continued to spread aggressively, with 12,571 cases reported by May, emphasizing the need for close monitoring and research into its characteristics and potential impact.

Correlation with Seasonal Trends

Examining the data, it is apparent that some variants, such as Alpha and Beta, did not exhibit a clear correlation with seasonal changes, as they peaked and declined independently of the time of year. In contrast, the Delta variant seemed to thrive during the summer months of 2021 but declined significantly in the following winter.

Omicron (BA.1) and Omicron (BA.5) displayed a different pattern, with their highest cases reported during the colder months of late 2021 and early 2022, possibly suggesting a correlation with seasonal changes.

Notably, Omicron (XBB.1.5) emerged in late 2022 and continued to surge throughout the winter of 2023, raising concerns about the potential for seasonal exacerbation.

In summary, the COVID-19 pandemic has witnessed the rise and fall of various variants, with Alpha, Beta, and Gamma disappearing, Delta exhibiting a protracted presence, and Omicron variants dominating for a time. The emergence of Omicron (XBB.1.5) highlights the dynamic nature of this virus.

Understanding the factors contributing to variant prevalence, including seasonality, is crucial for public health measures and vaccine strategies. Vigilant monitoring and research are imperative as we navigate the evolving landscape of COVID-19 variants.

Conclusion

To summarize, the COVID-19 pandemic has had a profound and lasting impact on the United States and the global community. The analysis has provided crucial insights into the virus's trajectory, vaccination efforts, and the toll it has taken on the nation.

A dynamic pattern has been observed in virus-related deaths, with COVID-19 reaching its high point in 2021, followed by declines in 2022 and 2023. The vaccination campaign saw its peak in 2021, with Pfizer leading the way. Significantly, an inverse correlation between deaths and vaccinations was evident, underscoring the life-saving potential of widespread vaccination campaigns.

Furthermore, a positive correlation between deaths and infection cases highlighted the imperative of controlling virus transmission. The observed seasonal spikes in death rates during the winter months emphasized the necessity for ongoing vigilance and preparedness during these periods.

The state-level analysis illuminated population density, vaccination coverage, infection rates, and mortality across different states, with California, Texas, Florida, New York, and Pennsylvania experiencing significant impacts.

Lastly, the emergence and decline of COVID-19 variants, including Alpha, Beta, Delta, and Omicron, each presenting its own set of challenges were examined. Understanding these variants and their seasonal trends is crucial for tailoring public health strategies and vaccination efforts.

As the battle against the pandemic persists, these insights will be a guide in optimizing future responses, emphasizing the critical importance of vaccination and comprehensive public health measures. The journey toward a post-pandemic world continues, marked by resilience and a commitment to learning from the past.

Reference

- ◆ <https://data.cdc.gov/Vaccinations/COVID-19-Vaccinations-in-the-United-States-Jurisdi/unsk-b7fc>
- ◆ <https://data.cdc.gov/Vaccinations/COVID-19-Vaccine-Distribution-Allocations-by-Juris/b7pe-5nws>
- ◆ <https://data.cdc.gov/Vaccinations/COVID-19-Vaccine-Distribution-Allocations-by-Juris/saz5-9hgg>
- ◆ [https://data.cdc.gov/NCHS/Provisional-Death-Counts-for-Influenza-Pneumonia-a/ynw2-4viq](https://data.cdc.gov/NCHS/Provisional-Death-Counts-for-Influenza-Pneumonia/ynw2-4viq)
- ◆ <https://data.cdc.gov/NCHS/Provisional-COVID-19-Death-Counts-by-Age-in-Years-/3apk-4u4f>
- ◆ <https://data.cdc.gov/Case-Surveillance/Weekly-United-States-COVID-19-Cases-and-Deaths-by-/pwn4-m3yp>
- ◆ https://covid.cdc.gov/covid-data-tracker/#vaccinations_vacc-people-booster-percent-pop5
- ◆ <https://www.census.gov/topics/population.html>
- ◆ <https://data.ers.usda.gov/reports.aspx?ID=17827>
- ◆ <https://ourworldindata.org/grapher/covid-variants-area?country=~USA>
- ◆ <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9339207>